Question 32

Correct

Mark 10.00 out of 10.00

Flag question

Write a C program to find the equilibrium index of an array. The equilibrium index is an index such that the sum of elements at lower indexes is equal to the sum of elements at higher indexes

For example:

Test	Input	Result
case 1	7 -7 1 5 2 -4 3 0	Equilibrium index found at index 3
case 2	6 -7 1 5 2 -4 3	Equilibrium index found at index 3
case 3	5 -7 1 5 2 -4	No equilibrium index found

Answer: (penalty regime: 0 %)

Reset answer

```c
#include <stdio_ext.h>

int main()
{
    //Read the size from user
    int size;
    scanf("%d",&size);
    int arr[size];

    //Read elements from user
    for(int i=0;i<size;i++)
    {
        scanf("%d",&arr[i]);
    }

    //Define your logic hereca
    int sum1=0,sum2=0;
    for(int i=0;i<size;i++)
    {
        sum1=0;
        sum2=0;
        for(int j=0;j<i;j++)
            sum1=sum1+arr[j];
        for(int k=i+1;k<size;k++)
            sum2=sum2+arr[k];
        if(sum1==sum2)
        {
            printf("Equilibrium index found at index %d",i);
            return 0;
        }
    }
    printf("No equilibrium index found");
}
```

	Test	Input	Expected	Got
✓	case 1	7 -7 1 5 2 -4 3 0	Equilibrium index found at index 3	Equilibrium index four
✓	case 2	6 -7 1 5 2 -4 3	Equilibrium index found at index 3	Equilibrium index four

Passed all tests! ✓

▸ **Show/hide question author's solution (C)**

Correct

Marks for this submission: 10.00/10.00.

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EMERTXE LMS

Question 37

Correct

Mark 10.00 out of 10.00

Flag question

Write a program to read and print an employee's detail using structure

For example:

Test	Input	Result
case 1	100 Ramesh Bangalore 10000 Married	Employee details are : Id - 100 Name - Ramesh Address - Bangalore Salary - 10000 Marital Status - Married
case 2	200 Dinesh Chennai 25000 Married	Employee details are : Id - 200 Name - Dinesh Address - Chennai Salary - 25000 Marital Status - Married

Answer: (penalty regime: 0 %)

Reset answer

```c
#include <stdio_ext.h>

// Define the structure for employee details
struct Employee
{
    int id;
    char name[25];
    char addr[40];
    int sal;
    char mstatus[15];
};


int main() {

    //Define your logic here
    struct Employee E1;
    scanf("%d",&E1.id);
    scanf("%s",E1.name);
    scanf("%s",E1.addr);
    scanf("%d",&E1.sal);
    scanf("%s",E1.mstatus);

    printf("Employee details are : \n");
    printf("Id - %d\nName - %s\nAddress - %s\nSalary - %d\nMarital Status - %s

    return 0;
}

```

	Test	Input	Expected	Got
✓	case 1	100 Ramesh Bangalore 10000 Married	Employee details are : Id - 100 Name - Ramesh Address - Bangalore Salary - 10000 Marital Status - Married	Employee deta Id - 100 Name - Ramesh Address - Ban Salary - 1000 Marital Statu
✓	case 2	200 Dinesh Chennai 25000 Married	Employee details are :	Employee deta

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Write a C program to find the sum of each row and each column in a 2D array and display the results.

Input

Enter the size of rows and elements: 3 3

Enter the row and col elements:

1 2 3

4 5 6

7 8 9

Output

Row sums: 6 15 24

Column sums: 12 15 18

For example:

Test	Input	Result
case 1:	3 3 1 2 3 4 5 6 7 8 9	Row sums: 6 15 24 Column sums: 12 15 18
case 2:	2 3 6 3 -5 2 -3 9	Row sums: 4 8 Column sums: 8 0 4

Answer: (penalty regime: 0 %)

Reset answer

```c
#include <stdio.h>
void calculateSums( int row, int col, int arr[row][col]); //Function prototype
int main()
{
   //read row and col
   int row,col;
   scanf("%d%d",&row,&col);
   int arr[row][col];
   //read the array elements
   for(int i=0;i<row;i++)
        for(int j=0;j<col;j++)
            scanf("%d",&arr[i][j]);

   //call the function
   calculateSums(row,col,arr);

    return 0;
}
void calculateSums(int row, int col, int arr[row][col])
{

    // Calculate row sums and column sums
    int sumr=0,sumc=0;
    printf("Row sums: ");
    for(int i=0;i<row;i++)
    {
        sumr=0;
        for(int j=0;j<col;j++)
            sumr=sumr+arr[i][j];
        printf("%d ",sumr);
    }
    printf("\nColumn sums: ");
    for(int i=0;i<col;i++)
    {
        sumc=0;
        for(int j=0;j<row;j++)
            sumc=sumc+arr[j][i];
        printf("%d ",sumc);
    }


    // Display row sums

    // Display column sums

}
```

Correct

Mark 10.00 out of 10.00

Flag question

Write a C program for find the facotrial of the given number by using recursion.

For example:

Test	Result
printf("%d",factorial(5));	120
printf("%d",factorial(3));	6

Answer: (penalty regime: 0 %)

Reset answer

```c
#include <stdio.h>
int factorial(int n)
{
    //Logic
    int fact=1;
    for(int i=1;i<=n;i++)
        fact=fact*i;
    return fact;
}
```

	Test	Expected	Got	
✓	printf("%d",factorial(5));	120	120	✓
✓	printf("%d",factorial(3));	6	6	✓

Passed all tests! ✓

▸ **Show/hide question author's solution (C)**

Correct

Marks for this submission: 10.00/10.00.

Question 34

Correct

Mark 10.00 out of 10.00

Flag question

Write a function in c to extract a substring from a given string.

Example :

Input : this is test string

position to start : 9, length of substring : 4

Output : test

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Correct

Mark 10.00 out of 10.00

Flag question

Write a function in c to extract a substring from a given string.

Example :

Input : this is test string

position to start : 9, length of substring : 4

Output : test

For example:

Test	Result
char str[] = "yes this is true"; extractSubstring(str, 13, 4);	true
char str[] = "Emertxe information technologies"; extractSubstring(str, 9, 6);	inform

Answer: (penalty regime: 0 %)

Reset answer

```c
#include <stdio.h>
#include <string.h>

void extractSubstring(char str[], int start, int length)
{
    //Define your logic here
    for(int i=0;str[i]!='\0';i++)
    {
        if(i==(start+length-1))
            str[i]='\0';
    }
    printf("%s",(str+start-1));
}
```

	Test	Expected	Got	
✓	char str[] = "this is test string"; extractSubstring(str, 9, 4);	test	test	✓
✓	char str[] = "yes this is true"; extractSubstring(str, 13, 4);	true	true	✓
✓	char str[] = "Emertxe information technologies"; extractSubstring(str, 9, 6);	inform	inform	✓
✓	char str[] = "Interview Preparation"; extractSubstring(str, 16, 5);	ratio	ratio	✓
✓	char str[] = "There are 8 planets in the solar system"; extractSubstring(str, 14, 3);	lan	lan	✓

Passed all tests! ✓

▸ **Show/hide question author's solution (C)**

Correct

Marks for this submission: 10.00/10.00.

write a program to print a decimal number in binary equivalent and reverse of a decimal number using recursion.

For example:

Test	Input	Result
case 1	10	Binary equivalent: 1010 Reversed number: 1

Answer: (penalty regime: 0 %)

Reset answer

```c
#include <stdio.h>
#include<stdio_ext.h>
int rev(int n)
{
    int rev=0;
    while(n)
    {
        rev=rev*10+n%10;
        n=n/10;
    }
    return rev;
}
int main()
{
    int num,flag=0;
    printf("Enter a decimal number: ");
    scanf("%d", &num);

    //logic
    __fpurge(stdout);
    //print binary form
    printf("Binary equivalent: ");
    for(int i=31;i>=0;i--)
    {
        if((num & (1<<i)))
        {
            printf("1");
            flag=1;
        }
        else if(flag==1)
            printf("0");
        else if(flag==0 && i==0)
            printf("0");
    }
    //print reversed number
    printf("\n");
    printf("Reversed number: %d",rev(num));
    return 0;
}
```

	Test	Input	Expected	Got	
✓	case 1	10	Binary equivalent: 1010 Reversed number: 1	Binary equivalent: 1010 Reversed number: 1	✓
✓	case 2:	123	Binary equivalent: 1111011 Reversed number: 321	Binary equivalent: 1111011 Reversed number: 321	✓

Passed all tests! ✓

▸ **Show/hide question author's solution (C)**

Correct

Marks for this submission: 10.00/10.00.

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Write a C program to find the equilibrium index of an array. The equilibrium index is an index such that the sum of elements at lower indexes is equal to the sum of elements at higher indexes

For example:

Test	Input	Result
case 1	7 -7 1 5 2 -4 3 0	Equilibrium index found at index 3
case 2	6 -7 1 5 2 -4 3	Equilibrium index found at index 3
case 3	5 -7 1 5 2 -4	No equilibrium index found

Answer: (penalty regime: 0 %)

Reset answer

```c
#include <stdio_ext.h>

int main()
{
    //Read the size from user
    int size;
    scanf("%d",&size);
    int arr[size];

    //Read elements from user
    for(int i=0;i<size;i++)
    {
        scanf("%d",&arr[i]);
    }

    //Define your logic hereca
    int sum1=0,sum2=0;
    for(int i=0;i<size;i++)
    {
        sum1=0;
        sum2=0;
        for(int j=0;j<i;j++)
            sum1=sum1+arr[j];
        for(int k=i+1;k<size;k++)
            sum2=sum2+arr[k];
        if(sum1==sum2)
        {
            printf("Equilibrium index found at index %d",i);
            return 0;
        }
    }
    printf("No equilibrium index found");
}
```

	Test	Input	Expected	Got
✓	case 1	7 -7 1 5 2 -4 3 0	Equilibrium index found at index 3	Equilibrium index four
✓	case 2	6 -7 1 5 2 -4 3	Equilibrium index found at index 3	Equilibrium index four

Passed all tests! ✓

▸ **Show/hide question author's solution (C)**

Correct

Marks for this submission: 10.00/10.00.

Question 33

Correct

Mark 10.00 out of 10.00

Flag question